FACULTY OF COMPUTING

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GROUP PROJECT : FLIGHT TICKET BOOKING SYSTEM

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1.0 OVERVIEW OF THE PROJECT

Travelers have encountered many issues when it comes to booking either manually or through existing online platforms, such as slow interfaces, lengthy booking processes, limited availability, high prices, unfriendly user experiences, and complicated payment methods. These issues are further exacerbated by the threat of scams that are becoming more rampant nowadays. In fact, according to the Thales Group, the threat of cybersecurity has become alarming, as the emergence of AI and organized syndicates has made the threat more serious for both the airline industry and traveler safety. (“The rising cyber”, 2024) Because as we transit toward the future world where digital will be an inseparable part of our lives, it makes this matter more important than ever. On 23 March, a cyberattack by a group of an organized syndicate on KLIA rendered the airport's online system inoperable. This has triggered an alarm about the safety of the online system. (Nizam, 2025) It's critical for the development of the airline online booking system to be greater than the current existing system, as the threat is evolving. While providing a seamless experience to both the customer and the company that operates the system. Hence, our objective is to form a system application that will fulfill all the requirements, but at the same time, it will not jeopardize the security. That is when Elvanta Tech discovers that it is inconvenient for the customer to go through all of these problems. Elvanta Tech finds an idea to create the Velaris Flight Ticket Booking System to improve flight ticket booking systems by finding the solution for all the problems faced by customers. This will give customers a better experience and be able to increase the sales.

The online booking platform has various features to arrange schedules and handle the bookings part. For flight ticket booking systems through online platforms, there are airlines and OTAs such as Kayak, Expedia, and Skyscanner. These platforms all offer online flight management, booking, and search services. The service provided, including flight search and selection, will feature real-time availability, as well as options for searching, comparing, filtering, and sorting. For booking and payment, the platform will offer features such as a streamlined booking process, secure payment processing, and support for multiple currencies. In terms of booking management, it will include functionalities for viewing booking details, making modifications, handling cancellations, and processing refunds. Additional services will also be available, including baggage information, loyalty program integration, and customer support.

Velaris Flight Ticket Booking System is developing an Airline Reservation System (ARS), a software commonly used by airlines to manage and store flight information, passenger data, and booking records. It serves as a centralized database for flight schedules, seat availability, fare details, and passenger profiles. This system helps to improve the process of searching, booking, and purchasing flight tickets online. Nowadays, most of the customers have high expectations towards the flight ticket booking system. This makes the other system very hard to cope with and leads to customer dissatisfaction with that system and reduces the sales. They want a convenient, simple, responsive, user-friendly, and fast response where they can access it 24 hours anywhere. This study aims to create an integrated system and a user-friendly interface that allows customers to easily access any flight information in real-time, manage their own bookings, and be able to complete their payment through online payments. Velaris will have administrators that have tools to manage all the flight schedules and are able to operate while monitoring all the bookings efficiently. Velaris' goal is to have an efficient operation, reduce all the administrative overhead, and enhance the overall travel experience for all the customers by implementing this system for Velaris.

2.0 PROBLEM STATEMENT

The problem statement for this project is to address and solve the limitations present in an existing flight ticket booking system, such as searching for flights, selecting available flights, and paying for the reservation. All of this discovery led to the creation of an improved airline reservation system through the Velaris Flight Ticket Booking System. The Velaris Flight Ticket Booking System aims to overcome issues such as:

1. **Complex user interfaces**

   Complicated user interfaces will result in a dashboard that loads with all the icons, making it impossible to jump to the next page and select a flight. Customers also faced confusion due to the complex and non-intuitive booking process, which lacked clear guidance and straightforward navigation. Additionally, the customer has a limited number of dates to select from. This caused them to become frustrated, which raised the number of booking errors. Additionally, it will result in a greater reliance on customer service, which raises operating expenses. Customers will not return to this system because of all these issues, as it is not user-friendly, particularly for elderly and busy working adults.

2. **Payment failures**

   The systems experienced payment failures, either declining the payment or loading the page until it timed out. Another problem is that even if the payment fails, the seat stays the same and prevents the customer from making another reservation for a flight ticket. Hence, the customer will lose their trust in their payment towards the system, and the system will lose all the potential sales.

3. **Payment and refund issues**

   When a customer has already made the payment and that particular payment is successful, but the system was loading or the internet was down, it will lead to an unsuccessful booking, which will cause the flight system to refund pages. There are also system timeouts that happen while the payment gateway is processing. Customers may also pay more than the original price in cases where the system incorrectly converts the currency. All of these problems, however, were not resolved quickly, and they were consistently postponed or said that refunds were either unsuccessful or not possible due to an inefficient system.

4. **Security vulnerabilities**

   Regarding system security, the system might not be up to date with the most recent version, which leads to outdated encryption protocols and a decreased level of trust. This can result in insecure Application Programming Interfaces (APIs) with weak authentication mechanisms, potentially leading to customer data breaches. The government may impose regulatory fines when a company fails to follow the law, especially in cases where customer data is leaked to unauthorized parties. As a result, customers may lose trust in the airline.

5. **Search and filtering inefficiencies**

   When customers have difficulty finding flights using the search function, it indicates that the system has a poor filtering mechanism for flight ticket availability, often providing delayed or inaccurate results. Furthermore, the absence of specific times or dates for flights makes it difficult for customers to find the best ticket options, which may cause them to abandon the platform and choose not to return.

6. **Language and localization failures**

   A flight ticket booking system should support multiple languages to be globally accessible, helping attract and retain customers from around the world. If the system lacks translation features, it can be difficult for customers to select flight times accurately. Additionally, if the system is not accessible 24/7 and does not support time zone handling, international customers may face challenges when trying to book flights at their convenience. As a result, we risk losing potential global customers.

3.0 PROPOSED SOLUTIONS

I. Feasibility Study

- Improving productivity for end users (customers and airport personnel).
- Streamlined workflows tend to greater end-user experience.
- A well-planned flight ticketing system from the front end and back end enables a smooth workflow.
- The flight ticketing system selection procedures naturally lead to a user-friendly interface (front end).
- Real-time monitoring shows the most accurate data directly.
- Continue to maintain a reliable and consistent connection to the flight ticketing system.

II. Element of feasibility

1. **Technical Feasibility**
    - Review the integration capabilities
    - Check the current system of the requested company is capable of handling the add-on process later.
    - Identify the resource that meets the expectations.

2. **Economic Feasibility**
    - Cost of estimate for the entire project, including system study, employees' studies, and software/hardware.
    - Forecast extra revenue within the flight ticketing system
    - Cost possible financial risks with the airline due to unexpected expenses.

3. **Operational Feasibility**
    - Evaluate end-user readiness and acceptance for an upgraded flight ticketing system.
    - Make backup preparations for the data.
    - Assess the impact on daily operations so as to analyze system possibilities.

III. Cost - Benefit Analysis (CBA)

COSTS		Year 0	Year 1	Year 2	Year 3	Year 4
Development Cost (One-time)						
Integration with Airline Systems	10000	**11000**				
E-ticketing Software License	8500	**9350**				
Consultant Ticketing System	19000	**20900**				
Training of Airline Personnel	18500	**20350**				
Total (Development Cost)		**61600**				
Production Cost						
System Hosting	3500		3850	4120	4408	4716
IS Support	16500		18150	19421	20780	22235
Maintenance and Technical Setup	2500		2750	2943	3148	3369
Annual Production Costs			**24750**	**26483**	**28336**	**30320**
(PRESENT VALUE)			**22500**	**21886**	**21289**	**20709**
ACCUMULATED COSTS			**84100**	**105986**	**127276**	**147985**
BENEFITS		**Year 0**	**Year 1**	**Year 2**	**Year 3**	**Year 4**
Labor Cost Reduction	88000	78000	70200	73710	77396	81265
(PRESENT VALUE)			**63818**	**60917**	**58148**	**55505**
ACCUMULATED BENEFITS			63818	124736	182884	238389
GAIN OR LOSS			-20282	18749	55608	90405
PROFITABLE INDEX		**1.46761**				

Figure 3.1: Cost-Benefit Analysis Velaris Booking Ticketing System

4.0 INFORMATION GATHERING PROCESS

There are various ways of gathering information, such as interactive methods and unobtrusive methods. In this project, we are using interactive methods where we use interviews to explore the user experience of booking online flight tickets and distribute questionnaires by using Google Forms to the public. We choose this method because we have direct interaction with individuals instead of unobtrusive methods where data is collected indirectly.

4.1 METHOD USED

The methods we used were interviews and questionnaires. We collected feedback on individuals' experiences with online flight ticket booking through interviews. For the questionnaire, we distributed it to the public to gather insights about their experiences with booking flight tickets online.

Interview Session

The interview structure that we used is the pyramid structure. We start the interview with closed questions and continue to open-ended questions until the end of the question. We interviewed 29-year-old Mr. Syafiq Iezzat Bin Sabri, who works in Johor Bahru and regularly travels back to his hometown by plane. He frequently purchases plane tickets on the website. This is the closed question that we ask before we start with open-ended questions.

Figure 4.1 Interviewing Mr Syafiq Iezzat

The following is a list of the interview questions, which are the open-ended questions, and Mr. Syafiq's responses regarding the online flight ticket booking process. It is divided into three sections, which is Section A, Section B, and Section C.

Section A: General Experience

1. Can you describe your most recent experience booking a flight online?
   I currently use Flight Scanner, Agoda, Traveloka, and other third-party websites. Any user-friendly website that can assist me with filtering. However, I will immediately visit the airline website if I did not receive the best deal through the third-party websites. Occasionally, the airline's website offers better offers and more promotions than websites run by third parties. In addition, I am concerned about the security of third-party websites, which occasionally lag and delay while awaiting approval of the airline ticket. Moreover, its security is incredibly inadequate. However, I will turn to a third party if I want a less expensive option. Since I have a better experience on airline websites, I will go there right away if I don't want to get a headache. In contrast to the current system, I hope the other flight ticket booking system will perform significantly better.

2. What type of platform or website did you use, and why did you choose it?
   There are two platforms that I will use, which are the airline website and Flightscanner. If I am looking for the cheapest option, I usually check Flightscanner first. It is great for comparing prices across different third-party websites, so I can easily find the best deals. But when I want peace of mind and a more secure booking process, I prefer to use the airline’s official website. It might cost a bit more, but I feel more confident knowing I am booking directly with the airline.

3. How would you rate the ease of use of that platform?
   So far, my experience using the airline’s website has been pretty good. I usually prefer using a PC over a mobile application because the application can be slow and lacks some features. The website is not perfect either, it is often cluttered with ads and too much information. You need to go from point A to point B until you reach the booking flight ticket process. You have to click through so many steps just to book a ticket, and

sometimes it gets frustrating and tiring.

Section B: Booking Process

1. Were there any parts of the booking process that you found confusing or frustrating?
   I hope when we come to the airline, they are more straightforward. Like third-party websites such as Flightscanner can directly scan available flight tickets and all the deal prices. Like airline websites, they take some time, and the buffering rate is lower compared to third-party websites. This is one of the difficulties that I, as a customer, face a lack of choices for the flight ticket date, but it is more secure as you can get the email confirmation from the airlines through email. Like third parties, we will be afraid that we did not get the flight ticket. Everything has pros and cons.

2. Did you feel confident that you were seeing the best flight options and prices?
   When it comes to promotions, it can get a bit tricky whether you are using the airline's own website or a third-party platform. It is not always clear how much you are actually saving. Sometimes the promotion only covers the flight ticket, but extras like baggage and meals are not included. Timing also plays a big role. Airlines often offer deals during off-peak seasons, but not during busy times like Chinese New Year. And even if there is a promotion, you usually have to book six months in advance; last-minute deals are pretty rare.

3. Were there any technical issues, such as slow loading, payment failure, or missing details?
   Definitely, there are technical issues. The airline websites will take forever to load because they are packed with too much stuff, such as ads, promotions, and partnerships with other companies. On the other hand, third-party sites like Flightscanner can feel a bit sketchy. Some of the deals look too good to be true, and it is hard to tell if they are secure or trustworthy. For security, there is also the issue of missing information. Airlines do not always clearly state what is included in the ticket, whether it covers baggage, meals, seat selection, or any other amenities. Often, you have to dig around and figure things out on

your own, which can be frustrating. That has been one of the biggest challenges I have faced.

4. Did you receive clear confirmation and booking details after completing your purchase?
   Yes, purchasing from airlines is very easy, and we can receive the confirmation quickly via email. There is customer service available if anything goes wrong. But when you go through third-party platforms, it can be even more stressful for me. You need to go through so many steps from registering until waiting for confirmation. The delay that you experience can make you anxious. It is not always about trust. I had once booked through Traveloka, and when I had a question on my mind, there was no person in charge that I could ask around. There is no customer service from Traveloka that can assist me with my question. I was about to fly to Hong Kong and still do not have any information about their office or customer service. When there is no customer service, all the matters are delayed and take a longer time to process. You have already made plans and set your itinerary, if you do not receive the ticket on time, it might ruin all of our plans. The more time needed to process my flight ticket, the more I start to worry that I might have been scammed.

Section C: Expectations and Suggestions

1. What features or functions do you value most when booking a flight online?
   The thing that I value the most when booking a flight online is a fast loading speed. When I am planning a trip, I will survey and check multiple dates and different destinations. Hence, I need a platform that can give quick results and be responsive to my desires. As we know, traveling requires a lot of planning, that is why I need to choose the best flight, date, and time to travel that can fit my busy schedule. Since I am not always available to travel on specific dates or times, a system that is efficient and fast can help me to schedule my trip plan ahead. It also makes it easier for me to compare all the options available, and I can choose the most suitable, affordable flight for me without wasting my time.

2. What do you think could be improved in the current systems you have used?

   The main thing that needs to be improved in the current system is the transparency, especially when it comes to promotions. The promotion has a lot of terms and conditions clauses where they do not explain in detail what the promotion is or what we will actually get through the promotion. I think another thing that needs to be improved is the security. I hope more third-party websites are being verified and certified by the airline system itself. Hence, we will feel secure using it. I hope that the airline system should be able to control the third-party websites that sell the flight tickets under their company name. Otherwise, the scammers will misuse the airline brands and trick the customer. My idea is the airlines and third-party websites can collaborate to give customers the best deals and good customer service.

3. If you had to design your ideal flight booking system, what would it look like?

   If I could design my ideal flight booking system, it would be the combination of airline websites and FlightScanner. The system will be able to scan all the flight tickets from the whole airline company worldwide. But honestly, I think the airline would not do that because they will lose money as they have to sell their competitor's ticket on their own platform. But I will be happy enough if the system can provide a simple UI, a faster loading rate, and a further explanation about the flight ticket details. I would also like there to be 24-hour customer service provided in the system. I would like the human itself to answer the question and not the robot of customer service. Therefore, we do not need to further search the website or forum to look for an answer. It would definitely give customers a clearer picture of the promotion, which can help the customer to choose based on their preferences and budgets. Lastly, the competition between airlines and third-party websites is a must to benefit the customer and to make the company always provide a better service for customers.

4. How important is mobile accessibility when booking a flight? Do you usually use a phone or a computer?

   When it comes to booking a flight, I prefer using a computer compared to the phone. This is because the computer has a large screen, which makes it easier for me to survey, screen, and compare different timings of flight tickets. It also has a faster loading rate and is less confusing. At the computer, it is easier to choose back and forward to choose different dates of flight, this is more convenient for me.

If I am using a mobile application to survey flight tickets, it can be laggy, limited, and harder to use as it is not straightforward. In the mobile application, it is very hard to switch between dates and check the detailed information, such as flight time, baggage allowance, and seat selection. It is also because of the small screen of the mobile. If a mobile application has a simple UI and is easy to access, I may consider using the mobile application. But currently, I would find using the system by using a computer more convenient.

Closing Question

Is there anything else you would like to share about your experience with flight booking systems?

Based on my previous experience , I have had some negative views when using the flight booking system to book my flight. I bought a flight ticket through a third-party system for a trip to Hong Kong. This is my first time using the third-party system, as I am on a budget, hence the reason why I book through the third-party system. I was a bit surprised when I received an email claiming it was the staff from the booking company because the email address is different from the booking company's trademark name. But what was most surprising to me is that the email was half Mandarin and half English. This triggered a red alert, and panic set in inside of me thinking that I was getting scammed by the third-party system. I froze and was dumbfounded about the action that I should take to resolve this matter. I buy it because it is cheap and the review said it is legit. Fortunately, after I approach customer service, they manage to resolve this matter, although it does take a longer time, which makes me nervous again. Overall, it was a hell of an experience. As for the airline booking system, it is much better, safer, and faster customer service response . But they do have some flaws that need to be addressed urgently, such as flight delays, flight ticket seats, expensive tickets, less choice of flight times, and being unable to trace my booking ID. When it comes to payment generally I have no issue since the banking system used by airline booking systems are more well built and regularly updated, plus they have strict requirements during payment. As long as you use common sense and are vigilant, there should be no problem. In the worst-case scenario, there is a warranty on the payment you made, and it is possible to recover the money, which is lacking if you use the third-party websites. There is no assurance that they are safe and secure, so you must exercise extreme caution.Overall, more improvements need to be made to make it easier for travelers to travel while getting the best deal.

<u>Questionnaire</u>

1. Occupation

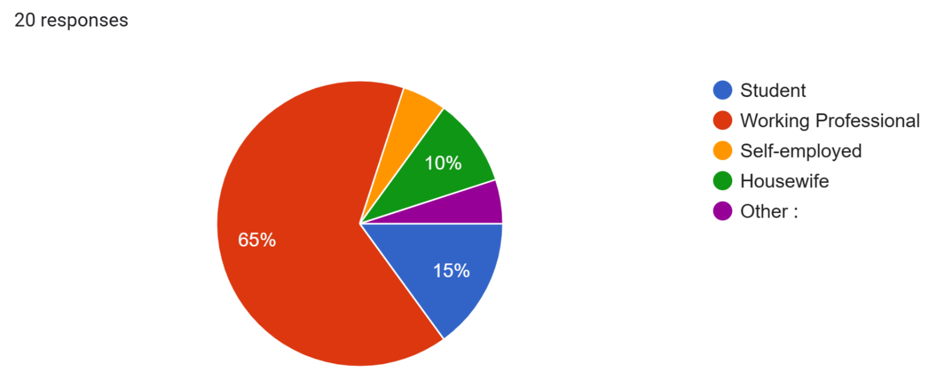


Figure 4.1.1 : Occupation Breakdown of Online Flight Booking Users

The survey on online flight ticket booking experiences reveals that 65% of respondents are working professional, followed by students (15%), housewife (10%), and self-employed/others (10%), highlighting the need for platforms to prioritize working professional with friendly features like discounts while also catering to business travelers with flexible booking options and family-oriented tools for occasional travelers.

2. Have you booked a flight online before?

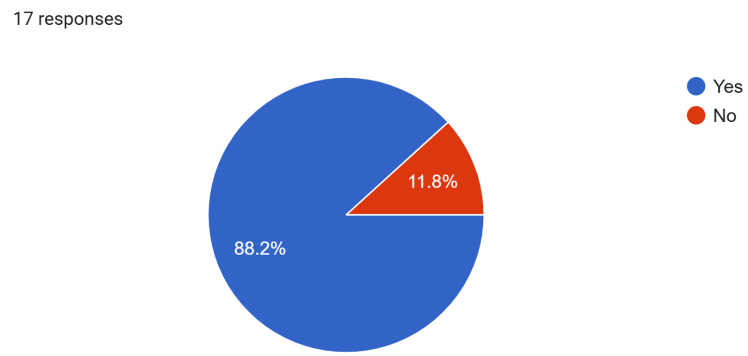


Figure 4.1.2 : Prevalence of Online Flight Reservation Methods

The survey reveals that 88.2% of respondents have booked flights online, demonstrating widespread adoption of digital platforms, while 11.8% remain first-time users, highlighting both the dominance of online booking and the ongoing need to support less tech-savvy travelers through user-friendly features and assistance options.

3. How often do you book flights online?

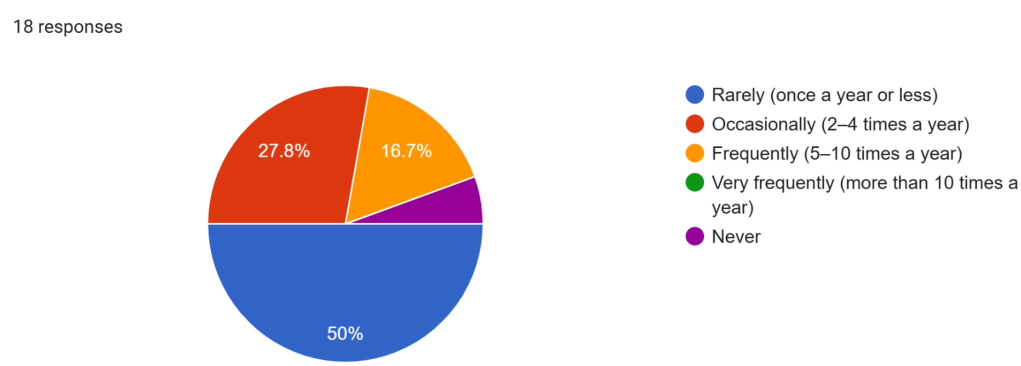


Figure 4.1.3 : Frequency Distribution of Online Flight Bookings

Survey data shows 50% of travelers book flights online rarely (once a year or less), as a need for us to attract more customers, while 27.8% occasionally (2 to 4 times a year) and 16.7%

frequently (5 to 10 times a year) represents there are high potential for flight ticket booking system still in needs and 5.5% never use the flight ticket booking system.

4. Which platforms have you used for booking?(Select all that apply)

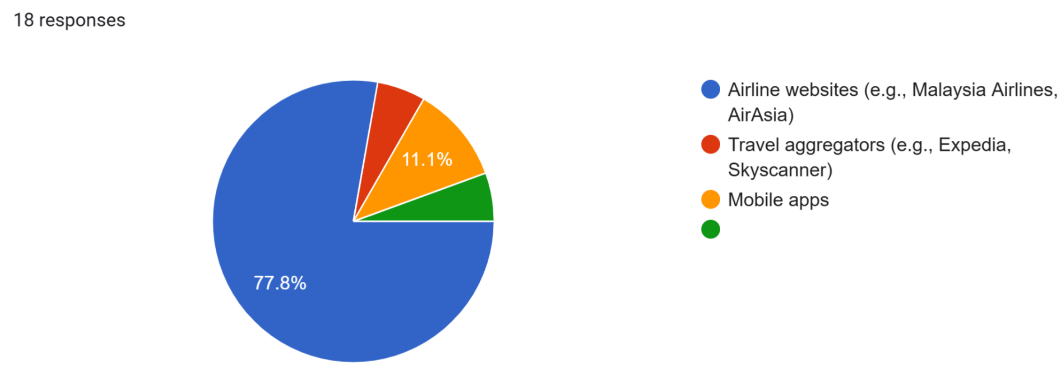


Figure 4.1.4 : Platform Preferences for Online Flight Bookings

The survey shows 77.8% of travelers prefer booking directly through airline websites, valuing their reliability and deals, while only 11.1% use aggregators (e.g., Expedia) and 11.1% use mobile apps. This 78-22 split suggests airlines should keep improving their websites' user experience, while third-party platforms need stronger unique features like better price comparisons or bundled offers to compete more effectively for the remaining market share.

5. What challenges have you faced during online flight booking?(Select all that apply)

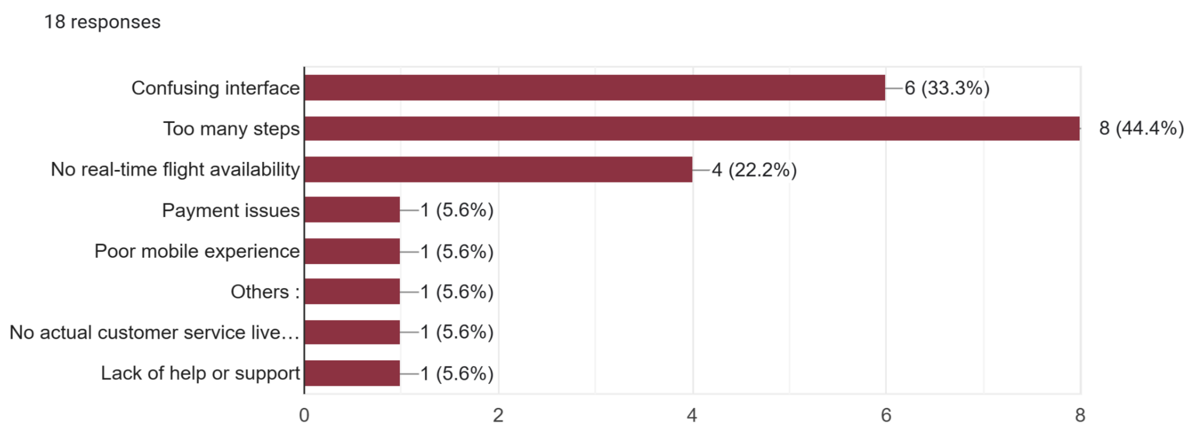


Figure 4.1.5 : Common Challenges in Online Flight Booking

This question reveals users' top online booking challenges: too many steps (44.4%), confusing interfaces (33.3%), and unreliable availability (22.2%). Less common but important issues include payment problems and poor support (5.6% each). These results indicate that streamlining the booking process and improving interface design should be top priorities for travel platforms, as usability issues outweigh technical problems in frustrating users. Addressing these pain points could significantly enhance the overall booking.

6. Have you ever been concerned about the security of your personal or payment information while booking a flight online?

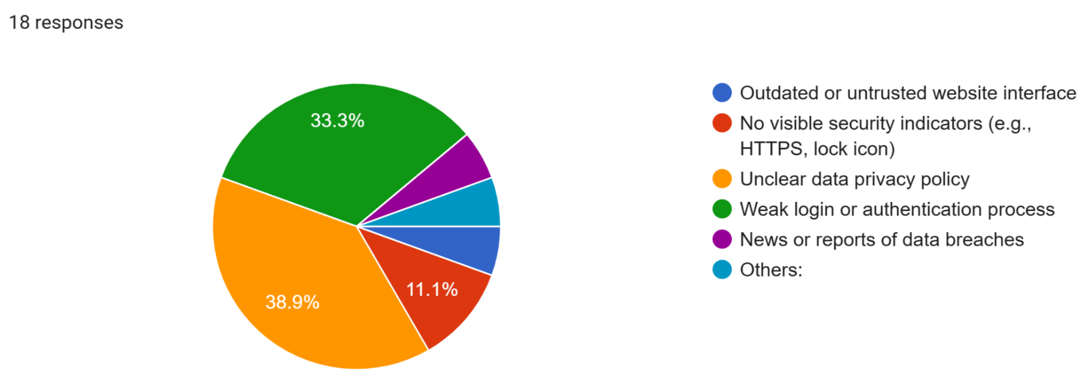


Figure 4.1.6 : User Security Concerns in Online Flight Booking

72% of flight bookings have security concerns, mainly about untrusted-looking interfaces (39%) or missing HTTPS indicators (33%). Only 11% worry about privacy policies. This shows users judge security by visual cues rather than technical details. Platforms should prominently display security badges, update outdated designs, and simplify privacy communications to build trust and reduce booking abandonment due to safety fears.

7. What issues have you experienced while searching or filtering flights on booking platforms?

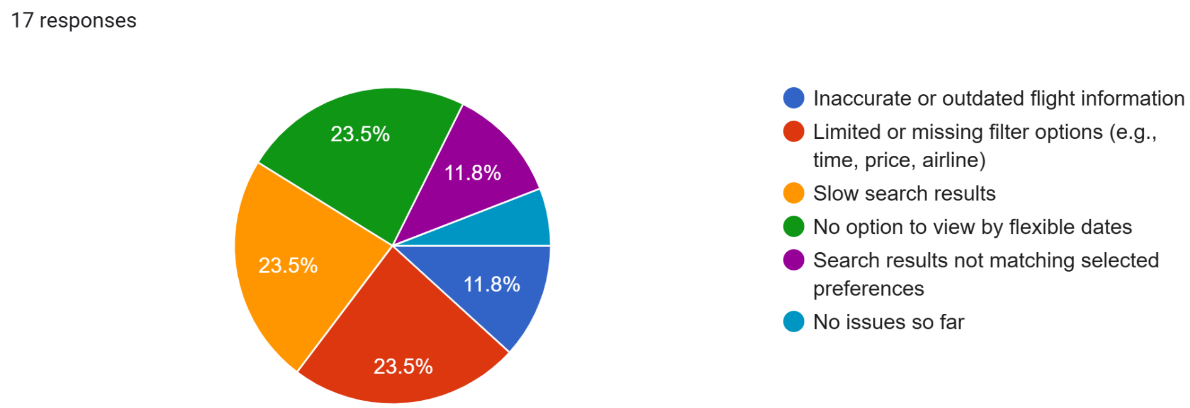


Figure 4.1.7 : User-Reported Issues in Online Flight Ticket Booking

The survey identifies three equally common flight search problems (23.5% each): no option to view by flexible dates, slow performance, and limited filters. Another 23.6% of users want better search results not matching selected preferences and inaccurate flight information. These findings suggest platforms should simultaneously improve data accuracy, expand filtering options, and optimize speed. With no single dominant issue, a balanced approach to these technical improvements would significantly enhance the search experience for most travelers while maintaining satisfaction for the 5.9% who report no current problems.

8. Which of the following would improve your experience with an online flight booking system? (Select all that apply)

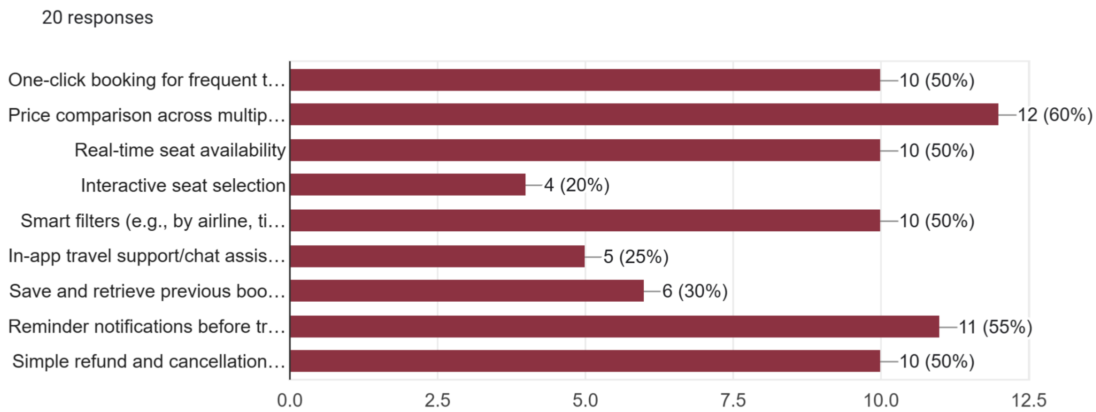


Figure 4.1.8 : Most Requested Improvements for Online Flight Booking Systems

The survey results indicate that users prioritize convenience, transparency, and efficiency in online flight booking systems, with price comparison across multiple airlines (60%) and reminder notifications before travel (55%) being the most requested improvements. Features like one-click booking, real-time seat availability, smart filters, and simple refund processes (each at 50%) also ranked highly, reflecting a demand for streamlined functionality. Lower-priority items, such as interactive seat selection (20%) and in-app chat support (25%), suggest these are less critical to the overall user experience. To enhance satisfaction, developers should focus on high-impact features like cost comparison and booking ease while considering niche enhancements based on further user feedback.

9.Which payment methods do you prefer when booking flights online?

(Select all that apply)

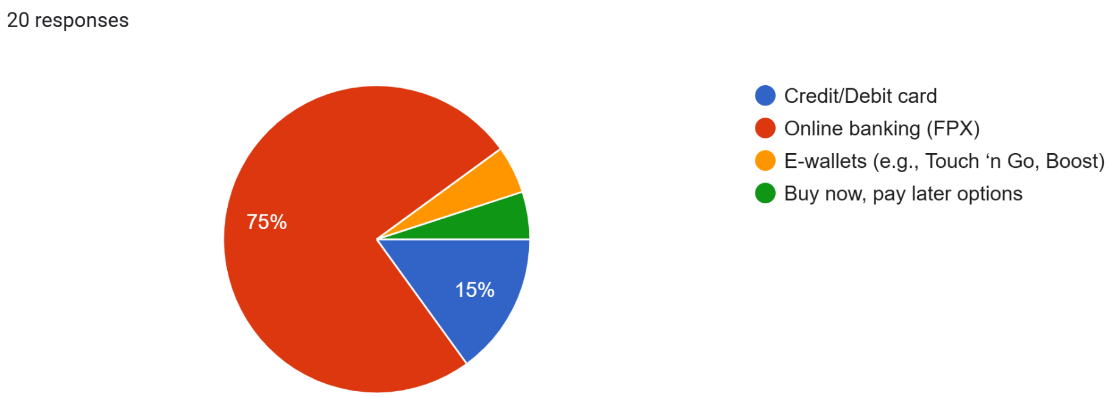


Figure 4.1.9 : Distribution of Payment Method

The results demonstrate a clear preference for traditional payment methods when booking flights online, with 75% of respondents opting for online banking. This overwhelming majority suggests that travelers continue to trust and value the security and convenience offered by online banking (FPX). This is due to the flexibility of payment, where they can pay anywhere and at any hour. In contrast, only 15% of users use credit or debit cards for payment. This may be due to the user needing to fill out more forms, and there are more steps to make it take longer to complete the payment. Digital payment methods like e-wallets (such as Touch 'n Go and Boost) collectively account for only 5% of preferences, indicating slower adoption in the flight booking sector compared to other e-commerce domains. The minimal usage of buy-now-pay-later options further reinforces travelers' inclination toward conventional payment solutions. These findings suggest that while airlines and booking platforms should certainly maintain and optimize their online banking payment systems, they may also want to invest in promoting digital payment alternatives to cater to evolving consumer behaviors, particularly among younger, tech-savvy demographics who are increasingly adopting these methods in other purchasing contexts.

10. What additional features would you like to see in an improved flight booking system?

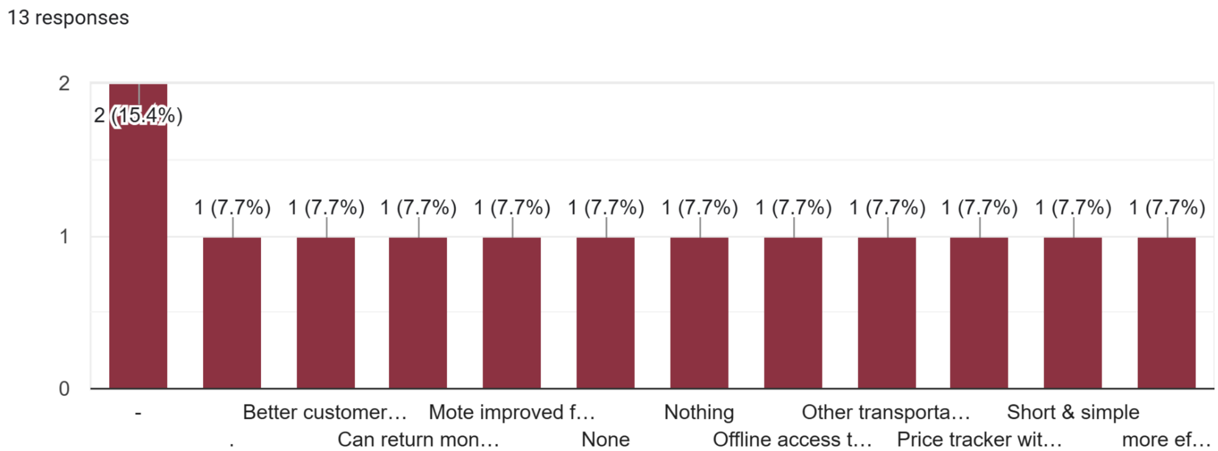


Figure 4.1.10 : User-Requested Additional Features for Flight Booking Systems

The survey results reveal that improved customer service is the most frequently requested enhancement for flight booking systems, mentioned by 15.4% of respondents. Beyond this primary concern, users identified several equally important but varied secondary features, each receiving 7.7% support: money-back guarantees, offline access, price tracking alerts, simplified booking processes, and multi-transport integration. Interestingly, 7.7% of respondents expressed satisfaction with current systems, requesting no changes. This distribution suggests that while customer service improvements should be prioritized, platforms have opportunities to test and implement specialized features that cater to niche needs. The diversity of responses indicates that no single advanced feature dominates user preferences, but rather that different travelers value different enhancements. Booking platforms should therefore focus first on strengthening core customer support capabilities while selectively piloting innovative features that address specific pain points for various user segments.

4.2 SUMMARY OF METHOD USED

In a nutshell, during the interview session, we find out that Mr. Syafiq Iezat is having a lot of difficulty booking tickets online, either in the airline system or in a third-party system. But if he needed to choose either one of the systems, he would choose the airline system for safety and ease of use. For the third-party system, he can filter a lot of choices from all types of airlines for a better option. He also found out that the systems need to be updated to a better version, as he did not find the current system user-friendly. He prefers to use a PC instead of other devices, as it has better viewing of the website interface and a faster loading rate. In comparison , when using system booking apps on a handheld device like a smartphone, the apps are always lagging and prone to crash under heavy usage. In addition, the ticket price increases exorbitantly during peak season, and the option of a refund is troublesome. To make matters worse, airlines usually offer credit points instead of cash. Plus, there are many ambiguities during the purchasing of the flight ticket, especially in information about the luggage size, seat selection, and hidden terms and conditions. He would like a better version of an online flight ticket booking system that is easy to access, cheap, has clear instructions, and can be accessed by using any device. The online booking system also needs to be streamlined from choosing the flight ticket until the payment process. Last but not least, a follow-up by customer service is highly appreciated.

The questionnaires were distributed to the public, and we got 20 respondents from a variety of age ranges. Most of the respondents are working professionals, and few of them are students, self-employed, or housewives. More than half of the respondents had booked a flight ticket through an online booking system. 50% of the respondents rarely book tickets online, and 16.7% frequently book tickets online. Most of the users prefer to use airline websites such as Malaysia Airlines and AirAsia compared to third-party websites such as Expedia and Skyscanner. The user's overall online booking experience was 72.2% good, which shows good improvement feedback from the user. The average overall online booking experience, according to 16.7% of respondents, is another reason why we require an enhanced flight ticket booking system. The challenges that 44% of users faced the most were too many steps in booking a flight online, and 33.3% of users were confused by the website interface. 33.3% of user was confused during the booking process, which proves that there are some users who still did not get clear instructions on how to book flight tickets online. 38.9% of the users had an unclear data privacy policy for personal and payment information. According to 33.3% of respondents, users are concerned about the website's inadequate login or authentication procedure. Most of the users had pointed out several issues, such as a lack of options in viewing flexible dates, slow search results, and limited or missing filter options during searching or filtering flights on the booking platform. 33.3% of users have difficulty understanding the flight ticket times due to time zone confusion, and it may lead them to miss out on the flight. For the user preference, 78.9% prefer a simplified booking process with fewer steps. 60% of the users would like to have price comparisons across multiple airlines to improve the experience of an online flight booking system. 40% of the users feel that it is very important to have a modern and visually clean interface when booking a flight. 70% of the users feel that they need the option to save their traveler details for faster future bookings. 75% of the users feel that they prefer online banking (FPX) when booking flights online. 52.65% of the users feel that they need to have flight delay or cancellation alerts for the update features. 40% of the users feel that they prefer a mobile app instead of a website for booking flights. One suggested improvement for the flight booking system is to enhance customer service, offer emergency refunds, streamline features and information, provide offline access to itineraries, include post-arrival transportation services, simplify the process, and add a price tracker with historical data and alerts for price drops, including a graph showing past trends for selected routes.

5.0 REQUIREMENT ANALYSIS

The Online Flight E-Ticket Booking System is designed to allow users to easily browse, select, and purchase airline tickets using a web or mobile application. The system enables users to view available flights based on input criteria such as origin, destination, travel dates, and number of passengers, and allows them to complete bookings with secure online payments.

The functional requirements include user registration and login flight. The airline system must validate each passenger's identification for the purpose of security and boarding. The airline companies are required to comply with aviation regulations, specifically the System for Advance Passenger Information (APIS), including the Secure Flight Program, Personal Data Protection Act (PDPA), and General Data Protection Regulation (GDPR).

Users shall find available tickets based on the flight schedule and make a booking on the selected seat until the completion of the booking process. The system is a standalone web application that interfaces with external airline databases and third-party APIs to retrieve real-time flight and pricing information.

Users shall insert additional information and choose optional service selections (luggage, seat selection, meal preference, insurance policy). The system shall need the user's name , email, and contact number to generate the official e-ticket and send the booking confirmation via email or SMS. The airline system shall have support booking management features such as viewing, modifying, or canceling bookings.

The non-functional requirements emphasize system performance, security, availability, usability, and scalability. The system must deliver the information fast and responsively. Ensuring the users do not experience unnecessary delay. The platform must be reliable, user-friendly, and responsive, with secure handling of sensitive data like payment and personal details. The payment confirmation process should be completed in a few seconds (approximately 5 seconds). The user interface (UI) should be easy to navigate for any level of user expertise. Key stakeholders include travelers (end-users), airline operators, payment gateway providers, and system administrators. The system should support multiple languages and be accessible to any level of user’s ability. The system architecture should integrate with airline databases and payment systems and utilize a robust backend with a scalable database to support high user traffic.

5.1 CURRENT BUSINESS PROCESS

SCENARIO 1 :

A family of Japanese—dad, mom, and kids, 5 years old and infant—living in Kasukabe. Dad has been promoted to factory manager and is bringing high profits to the company. Dad wants to spend their holiday in Koh Samui. His wife, Mrs Nohara has to search for information about the flight.

Mrs Nohara is searching for the flight that is friendly to infants and kids. For a long journey flight, consider more legroom and comfort, a night flight to coincide with the baby's sleep schedule, specific service for a bassinet, and assistance with infant care. She is also prefer her to use her local language while using the booking system.

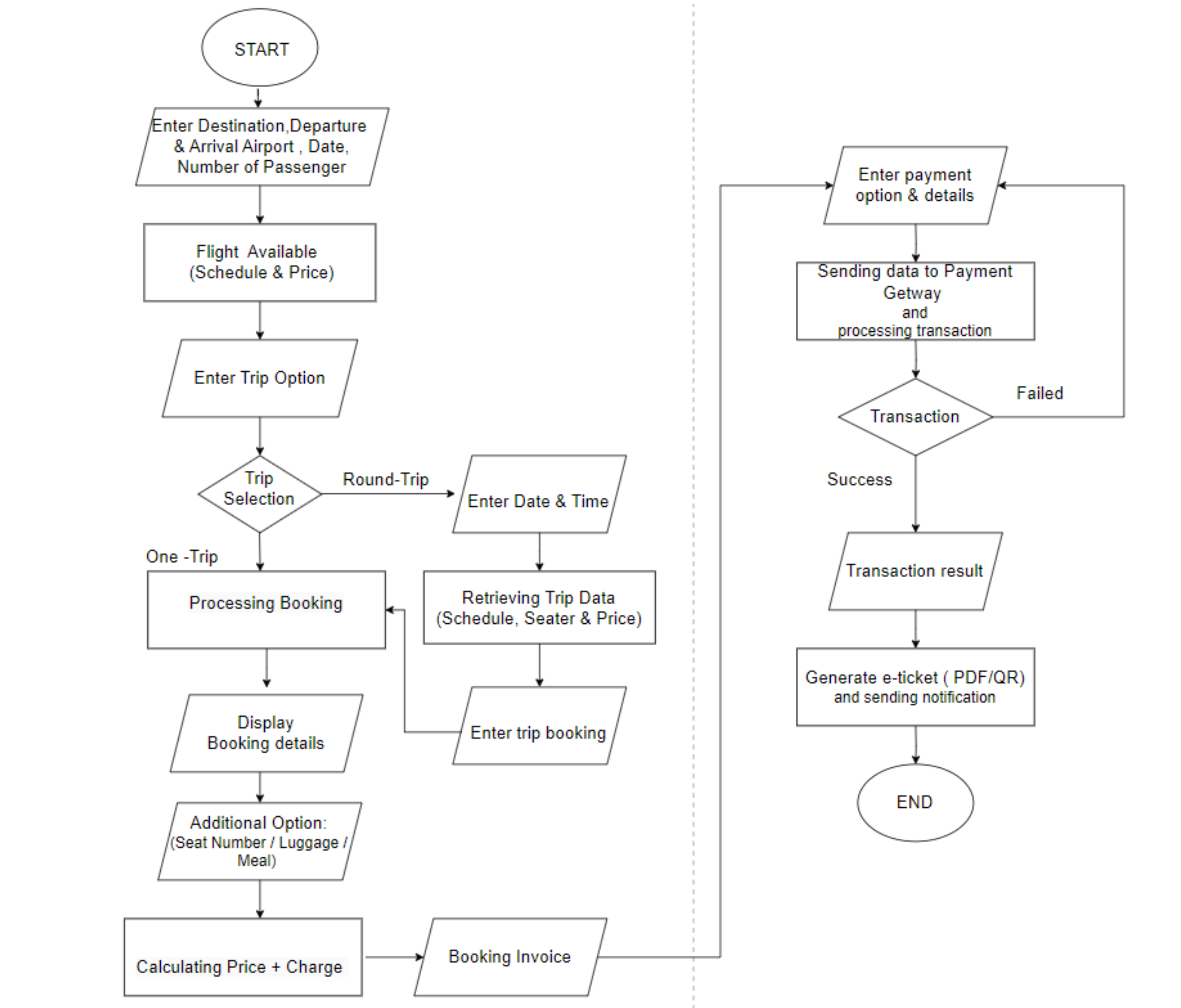
They never been in Thailand, so Mrs Nohara wants to know where they will be land at airport before heading to Koh Samui. Her son is a picky eater, she needs to choose what the meal served on the plane.

As a frugal housewife, Mrs Nohara seeks for the best affordable flight available.

UX shows an additional icon for the user to put in some requests. The system will search for suitable drop-down flight listings that match the user's preferences.

Scenario 1 has includes
Complex user interfaces (problem 1), search and filtering inefficiencies (problem 5), and language and localization failures (problem 6).

WORKFLOW


START
Enter Destination,Departure & Arrival Airport , Date, Number of Passenger
Flight Available (Schedule & Price)
Enter Trip Option
Trip Selection
Round-Trip
Enter Date & Time
One -Trip
Processing Booking
Retrieving Trip Data (Schedule, Seater & Price)
Display Booking details
Enter trip booking
Additional Option: (Seat Number / Luggage / Meal)
Calculating Price + Charge
Booking Invoice
Enter payment option & details
Sending data to Payment Getway and processing transaction
Transaction
Failed
Success
Transaction result
Generate e-ticket (PDF/QR) and sending notification
END

5.2 FUNCTIONAL REQUIREMENT

Functional requirements are statements of the services that the system must provide or are descriptions of how some computations must be carried out.

Requirement ID	Requirement Statement
FR001	The user interface shall be easy to navigate.
FR002	The system shall eliminate the duplicate entries.
FR003	The system shall develop auto-complete for destination and dates to save time in the booking process
FR004	The system shall lock the selected seat until the booking process is complete. (To prevent reserving the same flight seat)
FR005	The schedule must list available flights and suggestions with a 24-hour gap.
FR006	Users shall select the preferred language from the drop-down language preference. (If no language specified, switch to default language.)

Input-Process-Output Functional Requirement

FUNCTIONAL REQUIREMENTS

INPUT	PROCES	OUTPUT
Insert destination, date, number of passenger, departure, class type	System search available flight based on criteria	Listings of flight fare and schedule
Select preferred flight	Retrieve price details and policies Checking available seat Locking selected seat	Flight summary details
Insert information of passenger (name, Identity Number, Phone number)	Validate information Save temporary booking	Confirmation of passenger valid data Booking summary
Insert option (luggage, meal preference, seat selection, travel insurance	Update total cost and options	Additional booking summary
Select payment method	Processing payment through payment gateway	Payment confirmation, booking status update
Booking reference number Insert email	Generate booking reference Issue e-ticket	Display Booking summary, send notification email/sms

5. 3 NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements define how well the system should perform rather than what it should do. Non-functional requirements focus on the system's performance, stability, security, usability, scalability, and reliability.

Performance Requirements:

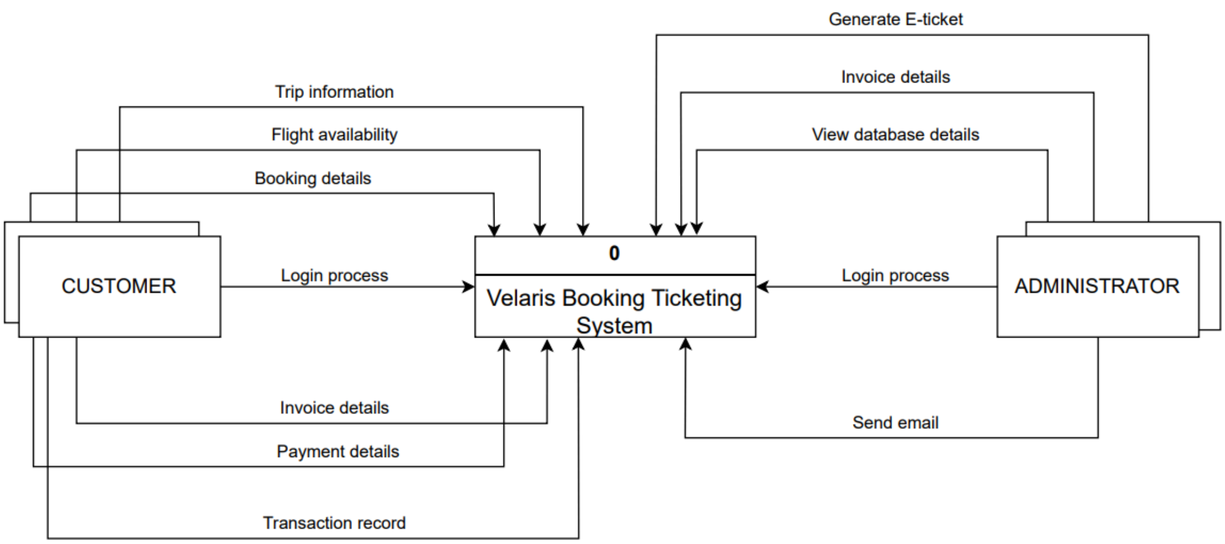
Requirement ID	**Requirement Statement**
NFR-P001	The system shall be responds to user query in 3 seconds
NFR-P002	The system should complete the booking process with in 10 seconds for normal rate.
NFR-P003	The system shall support for 10000 users without delay or disruption.
NFR-P004	The system should maintain 85% of normal performance during peak hours using.

Control Requirements:

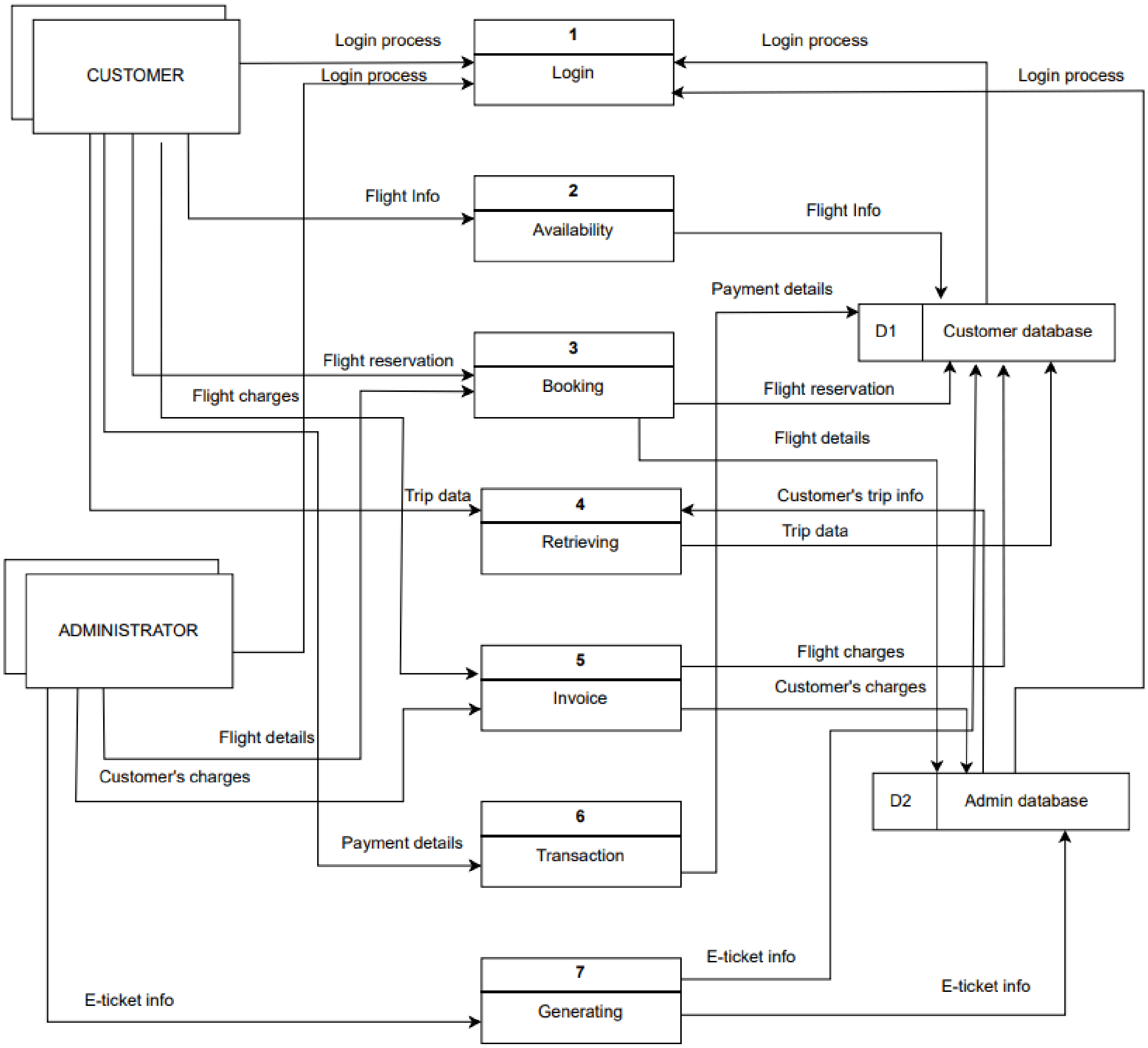
Requirement ID	**Requirement Statement**
NFR-C001	The system shall provide in real-time for administration tasks.
NFR-C002	The system admin shall control the booking services for maintenance purpose without effecting ongoing sessions or relevant data
NFR-C003	The booking session should automatically log out after 20 minutes of inactivity to prevent unauthorized use.
NFR-004	The system should be authorized to base control by admin to issue cancellation, refunds, and other control-level changes.

Logical DFD

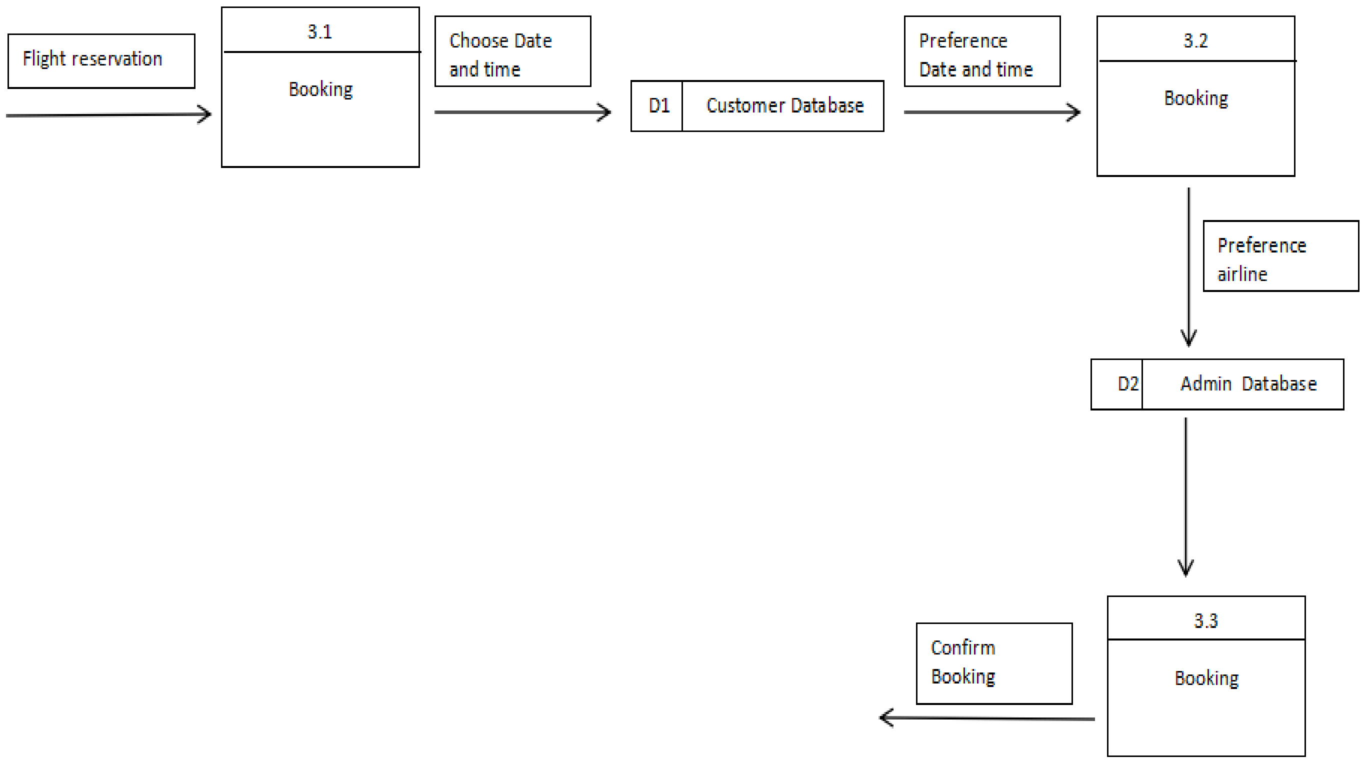
- Context Diagram



- Diagram 0


CUSTOMER
Login process
Login process
1
Login
Login process
Login process
Flight Info
2
Availability
Flight Info
Payment details
D1
Customer database
Flight reservation
3
Booking
Flight charges
Flight reservation
Flight details
Trip data
4
Retrieving
Customer's trip info
Trip data
ADMINISTRATOR
5
Invoice
Flight charges
Customer's charges
Flight details
Customer's charges
D2
Admin database
6
Transaction
Payment details
E-ticket info
7
Generating
E-ticket info
E-ticket info

- Child Diagram



6.0 SUMMARY OF REQUIREMENTS ANALYSIS

The Online Flight E-Ticket Booking System is made to give users an easy and safe way to search for, book, and manage airline tickets using websites or mobile apps. It includes important features like user login, searching for flights with filters (like where you're flying from, where you're going, and travel dates), entering passenger information, choosing extra services (like baggage, seats, and meals), secure payment options, and generating e-tickets, as well as options to change or cancel bookings. Non-functional requirements emphasize high performance, robust security for sensitive data, 24/7 availability, user-friendly interfaces, and scalability to handle peak traffic. Key stakeholders, including travelers, airline operators, payment gateways, and system administrators, require integration with airline databases and payment systems, supported by a scalable backend architecture to ensure reliability and efficiency, resulting in a comprehensive, secure, and user-centric e-ticketing solution. This solution aims to streamline the booking process, enhance customer satisfaction, and reduce operational costs for airline operators. By adopting the latest technologies and best practices in software development, it positions itself as a leading platform in the competitive travel industry.

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